

CS 473 (Spring 2025)

Homework 7 Solutions

Problem 7.1: Let $G = (V, E)$ be an unweighted graph with n vertices and m edges ($m \geq n$). A loop of length $\ell \geq 2$ refers to a walk $\langle v_1, v_2, \dots, v_\ell, v_1 \rangle$ such that $v_1, v_2, \dots, v_\ell$ are all distinct and $v_1v_2, v_2v_3, \dots, v_\ell v_1 \in E$. Note that a loop is the same as a (simple) cycle, except for the case of $\ell = 2$ (usually $\langle v_1, v_2, v_1 \rangle$ is not considered a cycle, but is a loop of length 2 if $v_1v_2 \in E$). We study the following problem: find a vertex-disjoint collection of loops and paths, maximizing their total length.

- (a) (20 pts) Show that there always exists an optimal solution using only loops of length 2 or odd lengths, and no paths. (So, we can redefine the problem without paths and the problem is the same.)
- (b) (10 pts) If G is bipartite, show that the optimal total length is the same as $2k^*$, where k^* is the maximum matching size. (So, the problem is equivalent to maximum matching in the bipartite case.)
- (c) (70 pts) Now consider the problem in the general case when G is not bipartite. Show that the problem can also be reduced to bipartite maximum matching and so can be solved in $O(mn)$ time by the Hungarian method or $O(m\sqrt{n})$ time by Hopcroft and Karp's algorithm.
(Hint: Construct a new bipartite graph G' where we create two versions of each vertex and two versions of each edge... Carefully prove correctness.)

Solution:

- (a) Consider an optimal solution S^* .
Suppose S^* contains an even-length loop $\langle v_1, v_2, \dots, v_\ell, v_1 \rangle$ of length ℓ . Then we can replace this loop by $\ell/2$ length-2 loops $\langle v_1, v_2, v_1 \rangle, \langle v_3, v_4, v_3 \rangle, \dots, \langle v_{\ell-1}, v_\ell, v_{\ell-1} \rangle$. Then the total length remains the same and so we still have an optimal solution.
Next suppose S^* contains a path $\langle v_1, v_2, \dots, v_{\ell+1} \rangle$ of length ℓ . Then we can replace this path by $\lceil \ell/2 \rceil$ length-2 loops $\langle v_1, v_2, v_1 \rangle, \langle v_3, v_4, v_3 \rangle, \dots, \langle v_{\lceil \ell/2 \rceil-1}, v_{\lceil \ell/2 \rceil}, v_{\lceil \ell/2 \rceil-1} \rangle$. Then the total length does not decrease (since $2 \lceil \ell/2 \rceil \geq \ell$) and so we still have an optimal solution.
- (b) A bipartite graph G does not have odd-length cycles. So, by (a), there exists an optimal solution S^* using only loops of length 2, and no paths. Let k be the number of such loops. The total length of S^* is $2k$. Since these k loops form a matching of size k , we have $k \leq k^*$. Thus, the total length of S^* is at most $2k^*$.
Conversely, given a maximum matching M^* of size k^* , we can map each edge uv in M^* to a loop $\langle u, v, u \rangle$ and obtain a collection of vertex-disjoint loops of total length $2k^*$. Thus, the total length of an optimal solution is at least $2k^*$.

(c) Construct a new bipartite graph $G' = (V', E')$, where

- $V' = \{(v, 1) : v \in V\} \cup \{(v, 2) : v \in V\}$.
- $E' = \bigcup_{uv \in E} \{(u, 1)(v, 2), (v, 1)(u, 2)\}$ (i.e., for each edge uv in G , we create two edges $(u, 1)(v, 2)$ and $(v, 1)(u, 2)$ in G').

Compute a maximum matching M^* in G' by the Hungarian method or Hopcroft and Karp's algorithm. Define the multiset $H = \{uv : (u, 1)(v, 2) \in M^*\} \cup \{uv : (v, 1)(u, 2) \in M^*\}$, which consists of $|M^*|$ edges (including multiplicity).

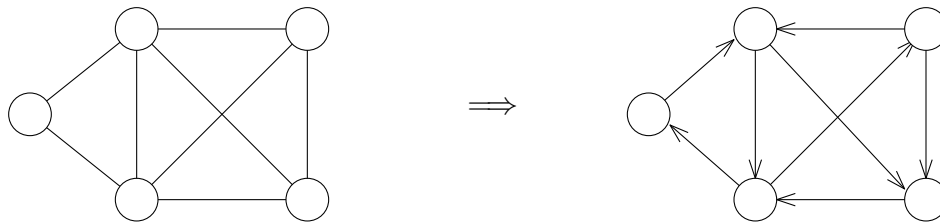
Observe that H is a multigraph with maximum degree at most 2 (since M^* has maximum degree 1). Some edges in H may have multiplicity 2; for these such edge uv , we add the length-2 loop $\langle u, v, u \rangle$ to the output S . The remaining edges forms a graph with maximum degree at most 2 and can be decomposed into vertex-disjoint cycles and paths (as observed in class); we add each these cycles and paths to the output S . The total length of S is exactly $|M^*|$. We return S .

To prove correctness, it suffices show that any optimal solution has length at most $|M^*|$. Let S^* be an optimal solution, of total length ℓ^* . By (a), we may assume that S^* consists only of loops. For each loop $\langle v_1, v_2, \dots, v_\ell, v_1 \rangle$, add $(v_1, 1)(v_2, 2)$, $(v_2, 1)(v_3, 2)$, $\dots$, $(v_{\ell-1}, 1)(v_\ell, 2)$, $(v_\ell, 1)(v_1, 2)$ to a set M . Then M is a matching in G' (since each vertex is used at most once), and the length of M is exactly ℓ^* . Since M^* is a maximum matching in G' , we have $\ell^* = |M| \leq |M^*|$.

(*Remark.* You don't need to know this, but this problem is related/equivalent to the notion of *fractional matching* or *half-integral matching*...)

Problem 7.2: Given an undirected graph $G = (V, E)$ and an integer d , we want determine whether it is possible to direct the edges so that the resulting directed graph has maximum **out-degree** at most d . Describe how to solve this problem by reduction to maximum flow. Prove correctness of your method.

Example: for the undirected graph below (left) and $d = 2$, the answer is yes, and one solution is shown on the right (there are many other solutions).



(Hint: start with a bipartite graph where the vertices on the left side are *edges* in G and the vertices on the right side are the vertices in G ; add a source and a sink...)

Solution:

Algorithm. First construct a directed graph (a flow network) $G' = (V', E')$, where $V' = \{s\} \cup E \cup V \cup \{t\}$ for two new vertices s (the source) and t (the sink), and E' is defined as follows:

1. For each $e \in E$, add the edge (s, e) to E' with capacity $c(s, e) = 1$.
2. For each $e = uv \in E$, add the edges (e, u) and (e, v) to E' , with capacity $c(e, u) = c(e, v) = 1$.
3. For each $u \in V$, add the edge (u, t) to E' with capacity $c(u, t) = d$.

Then compute the maximum flow f^* in G' , for example, by Ford and Fulkerson's method. Return yes iff $|f^*| = m$ (where $m = |E|$).

Running time. The graph G' has $O(m+n) = O(m)$ vertices and edges, and can be constructed in $O(m)$ time (as usual, we assume $m \geq \Omega(n)$). Ford and Fulkerson's method requires $O(m|f^*|)$ time, and since $|f^*| \leq \sum_{e \in E} c(s, e) = m$, we obtain an $O(m^2)$ -time algorithm. (We do not need Edmonds and Karp's variants here.)

Correctness: We claim that a solution to the problem exists $\iff |f^*| = m$.

Proof of ($\implies$): Suppose there is a directed version H of G with maximum out-degree at most d . We define a flow f as follows:

1. For each $e \in E$, set $f(s, e) = 1$.
2. For each $e = uv \in E$, if uv is directed as (u, v) in H , set $f(e, u) = 1$ and $f(e, v) = 0$, else set $f(e, u) = 0$ and $f(e, v) = 1$.
3. For each $u \in V$, set $f(u, t) = \text{out-deg}_H(u)$.

To verify that f is indeed a feasible flow, we check for

- capacity constraints:
 - For each $e \in E$, $f(s, e) \leq c(s, e)$: this is obvious ($1 \leq 1$).
 - For each $e = uv \in E$, $f(e, u) \leq c(e, u)$ and $f(e, v) \leq c(e, v)$: this is obvious since the left-hand side is 1 or 0 and the right-hand side is 1.
 - For each $u \in V$, $f(u, t) \leq c(u, t)$: this follows since $\text{out-deg}_H(u) \leq d$.
- flow conservation:
 - For each $e = uv \in E$, $f(s, e) = f(e, u) + f(e, v)$: this is obvious since the left-hand side is 1 and the right-hand side is $1 + 0$ or $0 + 1$.
 - For each $u \in V$, $\sum_{e=uv \in E} f(e, u) = f(u, t)$: this follows since the left-hand side is equal to the number of edges $uv \in E$ directed as (u, v) in H , i.e., $\text{out-deg}_H(u)$, and the right-hand side is $\text{out-deg}_H(u)$.

The value of the flow is $|f| = \sum_{e \in E} f(s, e) = m$. Thus, $|f^*| \geq |f| = m$. On the other hand, $|f^*| \leq \sum_{e \in E} c(s, e) = m$. We conclude that $|f^*| = m$.

Proof of ($\impliedby$): Suppose $|f^*| = m$. We define a directed version H of G as follows:

First note that $|f^*| = \sum_{e \in E} f^*(s, e) = m$ can happen only when all the terms have $f^*(s, e) = 1$. For each $e = uv \in E$, by flow conservation at e , $f^*(e, u) + f^*(e, v) = f^*(s, e) = 1$. By the *Integrality Theorem*, we know that $f^*(e, u)$ and $f^*(e, v)$ are integers. So, either $f^*(e, u) = 1$ and $f^*(e, v) = 0$, or $f^*(e, u) = 0$ and $f^*(e, v) = 1$. In the former case, direct e as (u, v) in H ; in the latter case, direct e as (v, u) in H .

For each $u \in V$, by flow conservation at u , $\sum_{e=uv \in E} f^*(e, u) = f^*(u, t)$. The left-hand side is equal to $\text{out-deg}_H(u)$. It follows that $\text{out-deg}_H(u) = f^*(u, t) \leq c^*(u, t) = d$. We have shown the existence of a directed version H of G with maximum out-degree at most d .

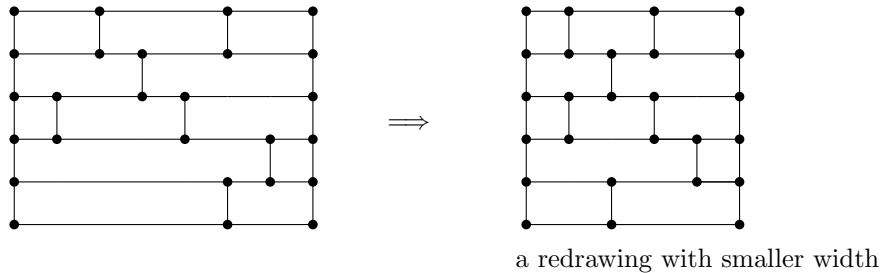
(Note: This problem is based on Asahiro, Jansson, Miyano, and Ono (IJFCS 2011).)

Problem 7.3: We are given a set of n axis-aligned rectangles $R = \{r_1, \dots, r_n\}$, where each rectangle r_i has a cost a_i . The rectangles r_i all have height 1 and have integer x - and y -coordinates, and no two rectangles intersect except along the boundaries, and their union is a rectangle $U(R)$. We want to redraw the rectangles as $R' = \{r'_1, \dots, r'_n\}$, whose union is a new rectangle $U(R')$, so that the rectangles r'_i still have height 1 and have integer x - and y -coordinates, and the adjacency structure is unchanged, i.e., if a side of the rectangle r_i touches a side of rectangle r_j , then the corresponding side of r'_i touches the corresponding side of r'_j .

Describe a polynomial-time algorithm to compute a redrawing S' that minimizes $\sum_{i=1}^n a_i \cdot \text{width}(r'_i)$, where $\text{width}(r'_i)$ denotes the width of r'_i .

You may assume that there is a polynomial-time algorithm for the following version of the *minimum-cost flow* problem: given a directed graph $G = (V, E)$ with source s and sink t and a value d , where each edge $e \in E$ is given numbers $\ell(e), c(e), \text{cost}(e)$, compute a flow f of value d , minimizing $\sum_{e \in E} \text{cost}(e) \cdot f(e)$, such that for every $e \in E$, we have $\ell(e) \leq f(e) \leq c(e)$, and we have conservation of flow at every vertex in $V - \{s, t\}$.

(Hint: create a vertex for each rectangle...)



Solution: Define a new rectangle $s = r_0$ whose upper side matches the bottom side of $U(R)$, and define a new rectangle $t = r_{n+1}$ whose lower side matches the upper side of $U(R)$. Let $a_0 = a_{n+1} = 0$. Let M be a sufficiently large number, e.g., $M = 2n$.

We construct the following directed graph (flow network) $G = (V, E)$:

- The vertices in V are the given rectangles, including s and t .
- For each pair of rectangles r_i and r_j such that the top side of r_i touches the bottom side of r_j , add an edge (r_i, r_j) with $\ell(r_i, r_j) = 1$, $c(r_i, r_j) = M$, and $\text{cost}(r_i, r_j) = a_j$.
- Add an extra edge (s, t) with $\ell(s, t) = 0$, $c(s, t) = M$, and $\text{cost}(s, t) = 0$.
- Define $d = M$.

The graph has $O(n)$ vertices. The graph has $O(n^2)$ edges and can be constructed in $O(n^2)$ time naively. (Actually, the number of edges is $O(n)$ and the construction time is $O(n \log n)$, by looking at each horizontal line, and sorting the rectangles from left to right per line. . .)

Compute a flow f^* with value d in this graph G minimizing the cost $C^* = \sum_{i,j} \text{cost}(r_i, r_j) \cdot f^*(r_i, r_j)$ by a polynomial-time algorithm for the minimum-cost flow problem.

Claim 1: We can produce a re-drawing with cost C^* .

Proof: Examine each horizontal line ℓ from bottom to top. Consider the sorted list of rectangles' corner points lying on ℓ . Examine the list from left to right. For each pair of consecutive elements (x, x') in this list (with x left of x'), suppose that the horizontal line segment from x to x' is bounded by rectangle r_i from below and rectangle r_j from above. Readjust the x -coordinate values so that the horizontal line segment from x to x' now has length equal to $f^*(r_i, r_j)$.

The order of the x -coordinates along each horizontal line is obviously preserved. By the Integrality Theorem for min-cost flows (which holds since all $\ell(e)$ and $c(e)$ values and d are integers), the coordinates are still integers.

Each rectangle $r_j \notin \{s, t\}$ is transformed into a rectangle r'_j in the re-drawing: the length of the bottom side of r'_j is equal to $\sum_i f^*(r_i, r_j)$, and the length of the top side of r'_j is equal to $\sum_k f^*(r_j, r_k)$, and by flow conservation, these two lengths must be equal.

Finally, $C^* = \sum_{i,j} \text{cost}(r_i, r_j) \cdot f^*(r_i, r_j) = \sum_{i,j} a_j f^*(r_i, r_j) = \sum_j a_j \cdot (\text{length of the top side of } r'_j) = \sum_j a_j \cdot \text{width}(r'_j)$, which by definition is the cost of the re-drawing. $\square$

Claim 2: Every re-drawing has cost at least C^* .

Proof: In the re-drawing, we may assume that every vertical line with integer x -coordinate intersecting $U(R')$ contains at least one corner point. If not, we can shrink the re-drawing and cost would be lowered.

Define a flow f as follows: for each pair of rectangles r_i and r_j (with $(r_i, r_j) \neq (s, t)$) such that the top side of r_i touches the bottom side of r_j , define $f(r_i, r_j)$ to be the length of the line segment formed by intersecting the boundaries of r'_i and r'_j . Define $f(s, t) = M - \sum_{r_j: (s, r_j) \in E, r_j \neq t} f(s, r_j)$. Then the value of the flow f is equal to $M = d$.

For $(r_i, r_j) \neq (s, t)$, we have $f(r_i, r_j) \geq 1 = \ell(r_i, r_j)$. Furthermore, by the above assumption, $f(r_i, r_j) \leq M = \ell(r_i, r_j)$. And we also have $f(s, t) \geq 0 = \ell(s, t)$ and $f(s, t) \leq M = \ell(s, t)$. For each $r_j \notin \{s, t\}$, $\sum_i f(r_i, r_j)$ is equal to the length of the bottom side of r_j , and $\sum_k f(r_j, r_k)$ is equal to the length of the top side of r_j ; since the two lengths are equal, we have flow conservation.

The cost of the flow f is $C = \sum_{i,j} \text{cost}(r_i, r_j) \cdot f(r_i, r_j) = \sum_{i,j} a_j f(r_i, r_j) = \sum_j a_j \cdot (\text{length of the top side of } r'_j) \sum_j a_j \cdot \text{width}(r'_j)$, which by definition is the cost of the re-drawing. Since C^* is the cost of the minimum-cost flow, $C \geq C^*$. $\square$

By these two claims, we obtain an optimal re-drawing.